```
    C22 = add(subtract(add(M1, M3), M2), M6)

    C = []

    for i in range(mid):
        C.append(C11[i] + C12[i])

    for i in range(mid):
        C.append(C21[i] + C22[i])

    return C


A = []
B = []

n = int(input())

for i in range(n):
    A.append(list(map(int, input().split())))

for i in range(n):
    B.append(list(map(int, input().split())))

result = strassen(A, B)

for row in result:
    print(*row)
```

```
4 5
6 5
7 8
58 57
59 60


...Program finished with exit code 0
```

```python
def next_power_of_two(n):
    p = 1

    while p < n:
        p = p * 2

    return p


def strassen_count(n):
    size = next_power_of_two(n)
    count = 1

    while size > 1:
        count = count * 7
        size = size // 2

    return count


def standard_count(n):
    return n * n * n


n = int(input())

print("Strassen Multiplications =", strassen_count(n))
print("Standard Multiplications =", standard_count(n))
```

```
4
Strassen Multiplications = 49
Standard Multiplications = 64


...Program finished with exit code 0
Press ENTER to exit console.
```

```
    AP = [[0] * size for _ in range(size)]
    BP = [[0] * size for _ in range(size)]

    for i in range(n):
        for j in range(n):
            AP[i][j] = A[i][j]
            BP[i][j] = B[i][j]

    CP = strassen(AP, BP)

    return [row[:n] for row in CP[:n]]


n = int(input())

A = []
B = []

for i in range(n):
    A.append(list(map(int, input().split())))

for i in range(n):
    B.append(list(map(int, input().split())))

result = multiply(A, B)

for row in result:
    print(*row)
```

input

```
7 8 9
30 36 42
66 81 96
102 126 150


...Program finished with exit code 0
Press ENTER to exit console.
```

```python
    C22 = add(subtract(add(M1, M3), M2), M6)

    C = []

    for i in range(mid):
        C.append(C11[i] + C12[i])

    for i in range(mid):
        C.append(C21[i] + C22[i])

    return C


n = int(input())

A = [[random.randint(1, 10) for j in range(n)] for i in range(n)]
B = [[random.randint(1, 10) for j in range(n)] for i in range(n)]

start = time.time()
standard_multiply(A, B)
standard_time = time.time() - start

start = time.time()
strassen(A, B)
strassen_time = time.time() - start

print("Standard Time =", standard_time)
print("Strassen Time =", strassen_time)
```

input

```
4
Standard Time = 3.1948089599609375e-05
Strassen Time = 0.00025010108947753906


...Program finished with exit code 0
Press ENTER to exit console.
```

```python
    C = []

    for i in range(mid):
        C.append(C11[i] + C12[i])

    for i in range(mid):
        C.append(C21[i] + C22[i])

    return C


n = int(input())
threshold = int(input())

A = []
B = []

for i in range(n):
    A.append(list(map(int, input().split())))

for i in range(n):
    B.append(list(map(int, input().split())))

result = strassen_hybrid(A, B, threshold)

for row in result:
    print(*row)
```

input

```
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16


...Program finished with exit code 0
Press ENTER to exit console.
```